

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

12

MARKING SCHEME

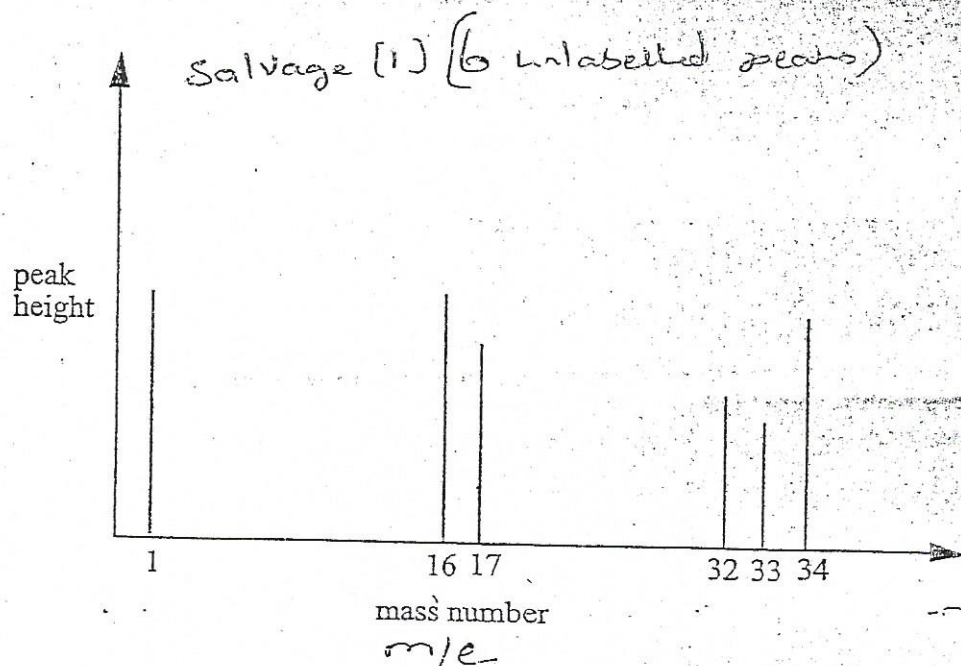
NOVEMBER 2013

CHEMISTRY 9189/1

1 (a) (i) $Ar = \frac{(63 \times 3.5) + (65 \times 1.5)}{5};$ [1]

$= 63.6;$ [1]

(ii)

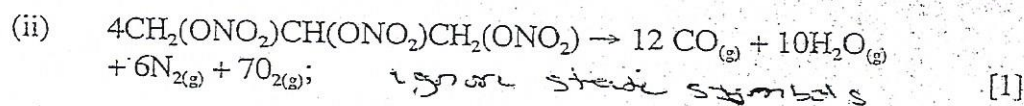


6 peaks (2)
3-5 (1)

Any 3 peaks; [1]

Total 4

(b) (i) HNO_3 ; nitric acid [1]



(iii) All gases produced are colourless/carbon soot not released; [1]

(iv) $RMM(\text{nitroglycerine}) = 227;$ [1]

$n(\text{nitroglycerine}) = \frac{1}{227} = 0.0044 \text{ moles};$ [1]

$n(\text{gas produced}) = 0.0044 \times \frac{35}{4} = 0.0385 \text{ moles};$ [1]

(v) $vol = 0.0385 \times 55 = 2.12 \text{ dm}^3;$ e.c.f. [1]

(vi) Poisonous CO is produced; or 12 [1]

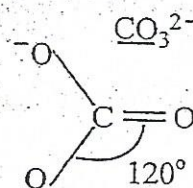
[Total 12]

(a) (i) Electron pairs, (whether bonded or not), arrange themselves in space so as to minimize mutual repulsions; [1] *AW*

- Repulsions are in the order: $lp - lp > lp - bp > bp - bp$; [1]

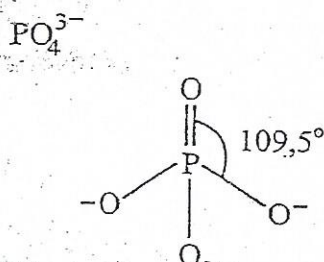
- Multiple bonds are considered as a single negative centre; [1] *AW*

(ii) $\underline{CO_3^{2-}}$ - (central atom - C) *A single bond*
 - number of bonded pairs - 3
 - number of lone pairs - 0
 - hence shape - trigonal planar; /



[1]

$\underline{PO_4^{3-}}$ - (central atom - P)
 - number of bonded pairs - 4
 - number of lone pairs - 0
 - hence shape - tetrahedral; /



[1] (5)

(b) (i) Molecules (pack together as efficiently as possible) held together by van der Waal's forces of attraction; / diag. of I_2 (as an example) [1]

(ii) An (array) of positively charged and negatively charged ions; / diag. of $NaCl$; *AW* *diag must be labelled*

[1]

(iii) An (array) of positive ions in a pool of delocalised electrons / diag.; [1] *AW*

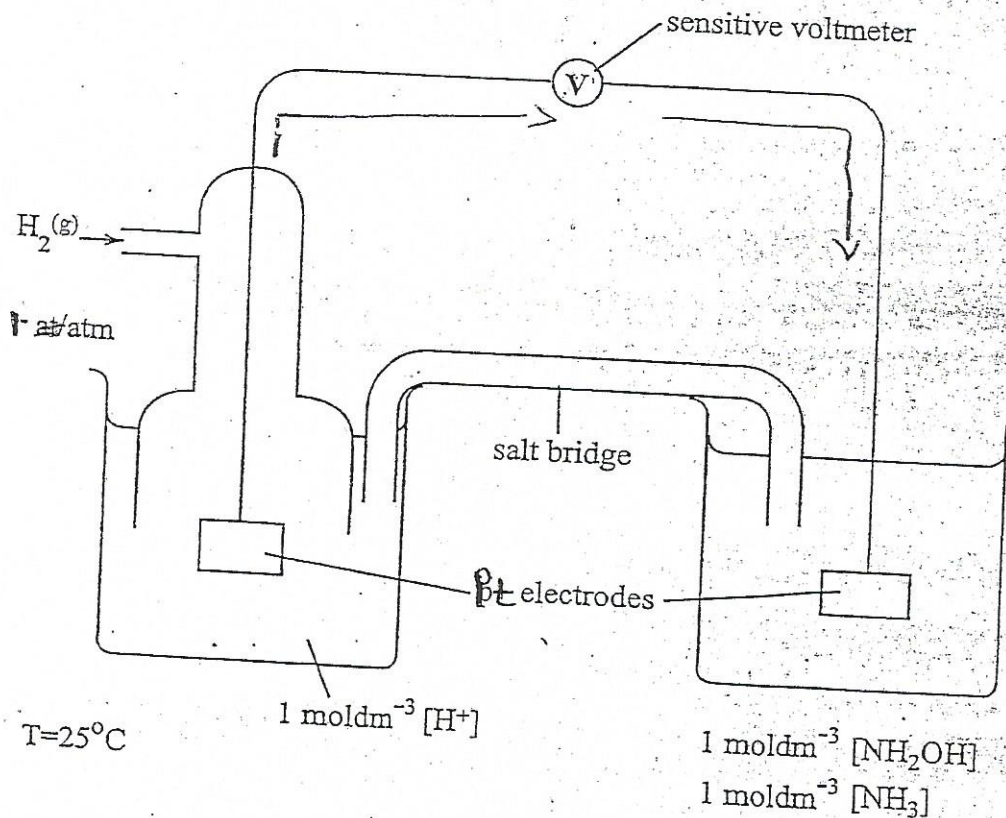
(c) - Ionic solids are *dissolve in* solvated by water but covalent solids do not interact with water; / *simple molecular*
ionic solids dissolve in water but simple molecular solids are insoluble
 Ionic solids conduct electricity in molten or solution form but covalent solids do not; *(aq) solution*
simple molecular [1]

[1]

- (d) 1. Stress may shift ions leading to alignment of similarly charged ions hence repulsions; / AW1 labelled diagram [1]
2. On application of stress, ions slide over each other (but still remain held together by the delocalised electrons); [1]
- [Total: 12]

- 3 (a) (i) Oxidation is increase in oxidation number or state
reduction is decrease in oxidation number; or state [1] *for both*
- (ii) Oxidation is loss of electrons
reduction is gain of electrons; [1] *for both*
- (b) (i) reduction $\text{NH}_2\text{OH} + \text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{NH}_3 + 2\text{OH}^-$; [1]
- oxidation; $\text{NH}_2\text{OH} + 2\text{H}_2\text{O} \rightleftharpoons \text{HNO}_3 + 6\text{e}^- + 6\text{H}^+$; [1]

ii)



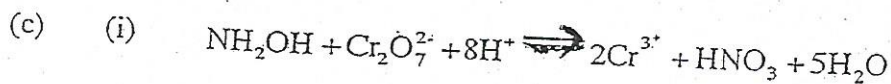
Correct diag; [1]
Standard Conditions; [1]
Any two labels; [1]
(total [3] for labels)

correct diag.; [1]
standard contias; [1]
Any two labels; [1]
(total [3] for labels)

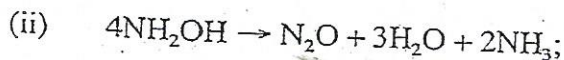
7 labels 3 mark
4 to 6 labels 2 marks
3 labels 1 mark

(iii) direction of flow of electrons S.H.E $\rightarrow$ NH_2OH sys.
(shown on diag.)

[1]



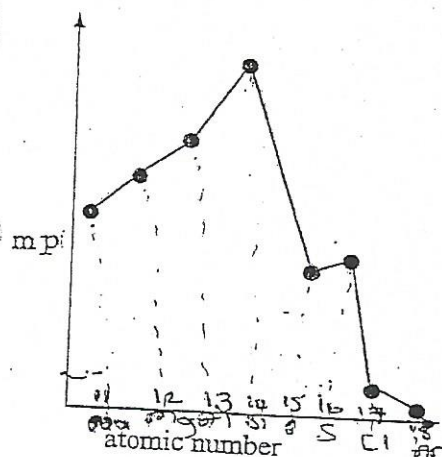
[1]



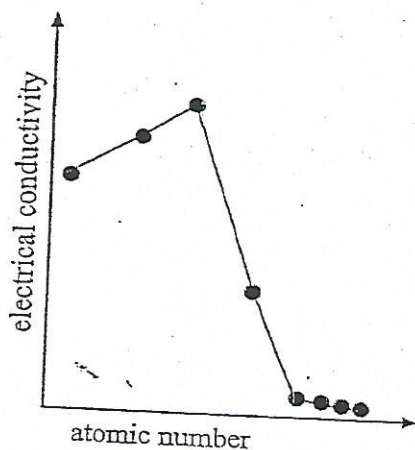
[1]

[Total: 12]

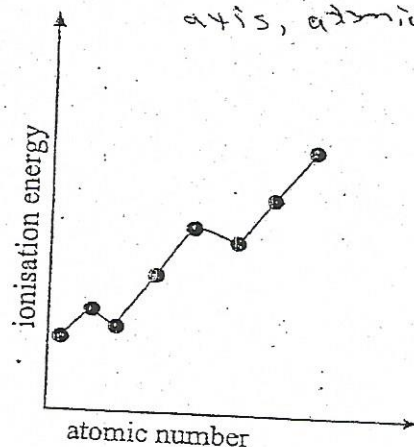
4 (a)



[1]



[1]



[1]

For the horizontal axis, atomic no/ele

Explain one
of horizontal
axis not
labelled

Increase from Na to Al is due to increase in strength of metallic bonds with increase in number of delocalised electrons;

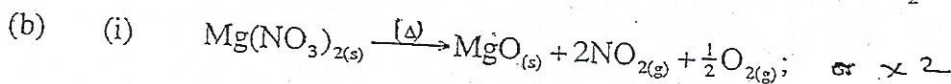
[1]

Si is giant covalent/macromolecular hence more energy is required to break strong covalent bonds; Al is

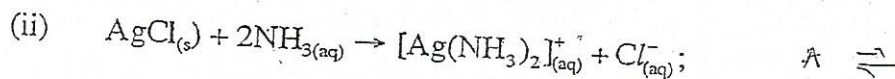
[1]

P to Ar are all simple molecular held together by weak van der waal's forces which increase in the order $\text{S}_8 > \text{P}_4 > \text{Cl}_2 > \text{Ar}$;

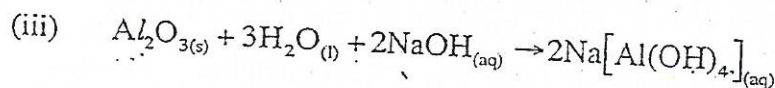
[1]



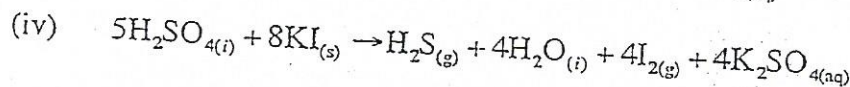
[1]



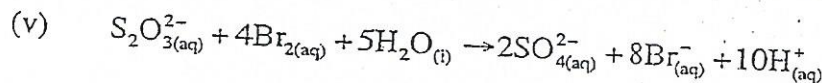
[1]



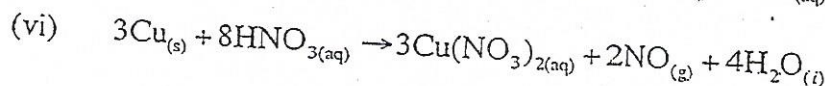
[1]



[1]



[1]

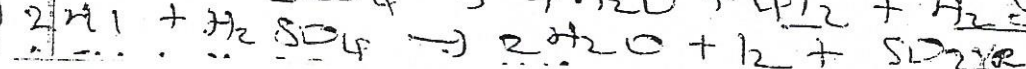
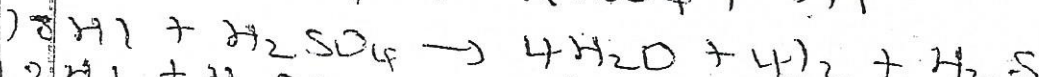
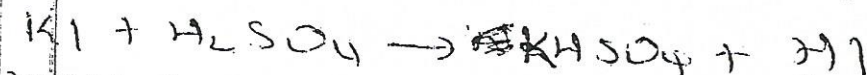


[1]

R ionic

[Total: 12]

State symbols if given should be correct.



N) 3 marks
Nogaw &
H₂S.

- 5 (a) Decrease; due to decrease in strength of metallic bonds as a result of decrease in charge density of metal ions in the metallic lattice; [2]
- (b) (i) (Neutralises acids) e.g milk of magnesia/ in tooth pastes; [1]
- (ii) Mild laxative/ ^{relief} constipation; [1]
- (iii) Refractory material / inner linings of furnaces; [1]
- (c) MgSO_4 - draws water from the boil; inflammation. [1]
- ethanol - kills the microorganisms; / bacteria [1]
- (d) Mix a little of the sample with water / acid; [1] *litmus is*
- Ca gives a solution that turns red litmus blue but no observable change [1] *for H₂O*
- with Mg / slower evolution of gas with Mg/AW; [1] *only*
- / burn a little of the sample; Mg burns with a dazzling white flame, Ca with a brick red flame; *if uses symbol both eqs and 2nd is cancelled*
- $\text{M} + 2\text{H}_2\text{O} \rightarrow \text{M}(\text{OH})_2 + \text{H}_{2(g)}$ / $\text{M} + 2\text{H}^+ \rightarrow \text{M}^{2+} + \text{H}_{2(g)}$ / $\text{M} + \frac{1}{2}\text{O}_2 \rightarrow \text{MO}$; [1] *if uses symbol both eqs and 2nd is cancelled*
- (e) Be^{2+} has the (smallest ionic radius hence) highest charge density; [1] *if uses symbol both eqs and 2nd is cancelled*
- It thus polarises the anion to give a high covalent character in the bonds; [1] *if uses symbol both eqs and 2nd is cancelled*
- high polarising power* [Total: 12]

- 6 (a) (i) An atom or a group of atoms substituting a hydrogen atom in a hydrocarbon; A/W an atom or group of atoms which changes the chemical properties of an organic molecule [1]
- (b) (i) $-\text{NH}_2$ - Makes the ring more reactive; [1] *2, 4 (6)*
- causes further substitution on ortho and para positions; [1] *ortho, para*
- (ii) $-\text{CHO}$ - Makes the ring less reactive; [1] *3, 5*
- causes further substitution on meta position; [1] *ortho electron donating*
- (c) phenyl amine - add aqueous Br_2 ; decolourised / white ppt; [2]
- carboxyl - Add PCl_5 / SOCl_2 ; white fumes of HCl ; [2]
- / Add $\text{Na}_{(s)}$; Effervescence (of $\text{H}_2(g)$); / gas bubbles
- / Add Na_2CO_3 ; Effervescence (of $\text{CO}_{2(g)}$); / gas bubbles
- Ref. heat is named alcohol, H₂SO₄ conc; Sweet smell*
- Keto - Add 2,4 dinitrophenyl hydrazine; orange ppt; (little acid catalyst) [2] *A yellow ppt*
- All reactions under ordinary conditions / r.t.p; [1]
- [Total: 12]

Be has very high 1st + 2nd IE which makes ionisation difficult; it shares e⁻ to form covalent compounds.

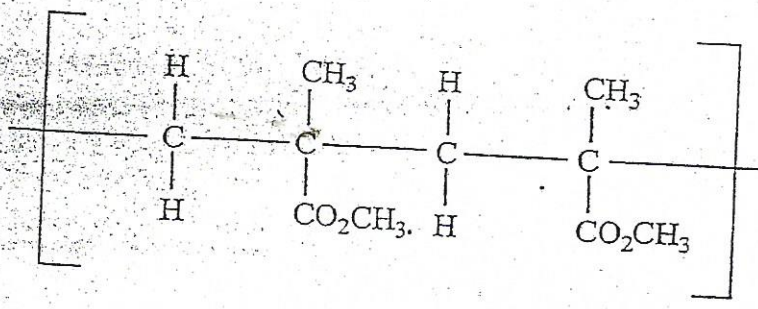
(a) (i) Methyl-2-methyl propenoate;

[1]

(ii) Addition polymer;

[1]

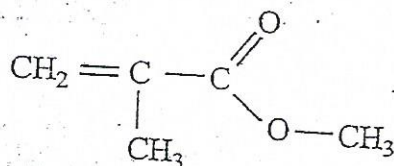
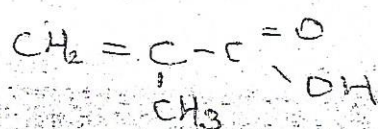
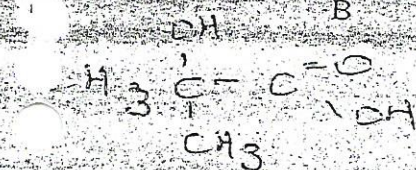
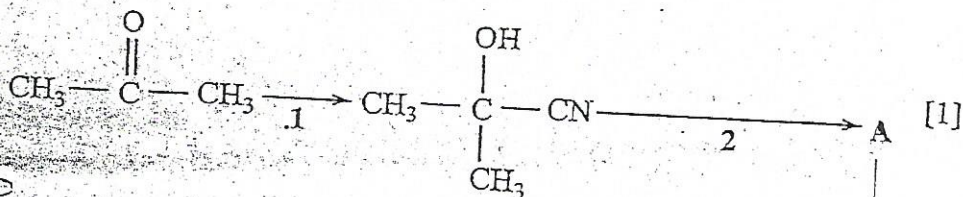
(iii)



3 repeat units only

[1]

(b) (i) A



3

4

B [1]

(ii) 1

HCN + (trace) NaCN / NaOH;

2

Heat with dilute H_2SO_4 / HCl;

3

Conc H_2SO_4 at 170°C / (Conc) H_3PO_4 at 300°C / hot Al_2O_3 ;

[1] A alkali

[1]

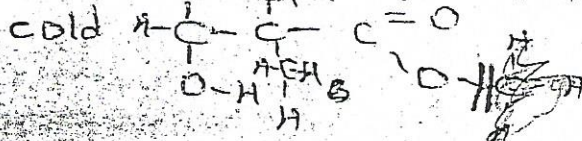
[1]

4

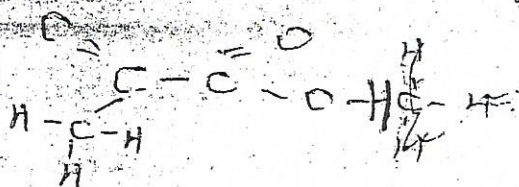
CH_3OH ; (a little) conc H_2SO_4 reflux;

[2]

(iii)



A is O-H is not displayed.

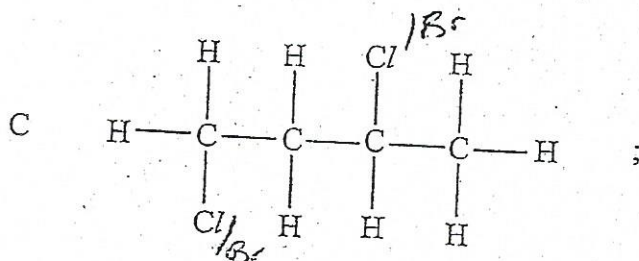
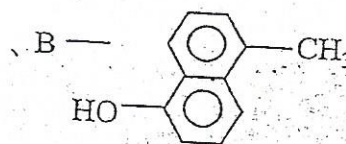
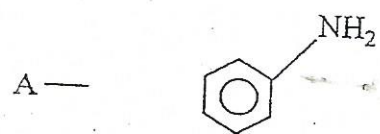


[1]

(iv) One of the carbon atoms on the double bond is attached to two hydrogen atoms / AW; NO chiral carbon

[1]

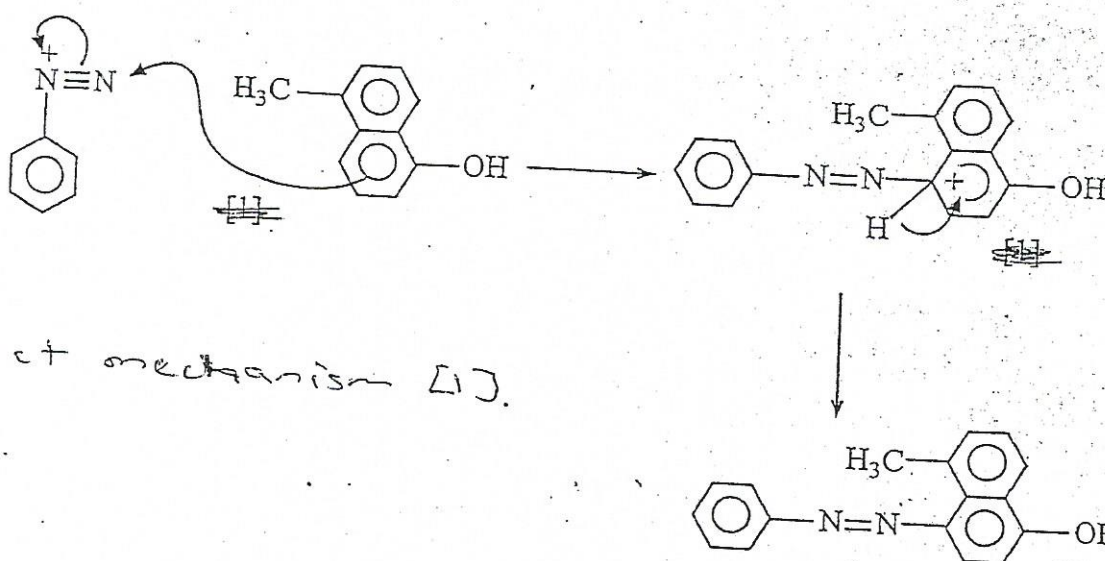
[Total: 12]



3

- (b) I - *Reflux, H₂O + NaOH (aq), hot conc KMnO₄ + H₂SO₄; [2]*
 - Hot concentrated KMnO₄ + H₂SO₄ / NaOH;
 II - *Mild alkali / NaOH at T < 10°C;* [1]
 IV - *(Conc) NH₃ heat under pressure / AW;* [1]
 (c) I - Oxidation; hydrolysis; [2]
 II - Diazotisation; [1]
 V - Nucleophilic substitution; [1]

(d)



[Total: 12]